GCI Final Project

〜Delivering AI-Powered Churn Prediction Solutions to Enhance Customer Retention〜

Introduction

Introduction & Project Scope
Introduction
This business proposal presents an AI-powered churn prediction solution for **Company A**, built on historical data to enhance customer retention. The project delivers a professional and actionable **Proof of Concept (PoC)** under clearly defined assumptions to ensure smooth execution.
Assumptions

Currency Standardization:
All monetary values (e.g., Monthly Revenue, Overage Charges) are standardized to **Dollars** for consistency. While absolute conversions may vary, the **ratios and relationships** between financial metrics remain unchanged.
Data Integration:
• Features from both **Client.csv** (customer information) and **Record.csv** (usage history) are integrated to enable the model to capture complex patterns linking **device age**, **billing spikes**, and **service quality**.
•**Handling Ambiguity:**
In line with Company A's data-protection policies, ambiguous features were refined through **advanced feature engineering** (e.g., *Overage Strain Ratio*) to ensure insights are **testable and business-relevant**.

Overview of the new AI Business

Three core business proposals

①Churn customer prediction

1. AI identifies customers who are at high risk of churn based on handset age and device value and other features.
2. By analyzing the characteristics of customers who have previously discontinued service, the model detects users likely about to churn

②Churn Customer Trend Analysis

1. By analyzing the behavior of customers who have previously left the service, the model uncovers engagement gaps and supports personalized retention campaigns, including tailored bundles, bonus usage incentives, and targeted marketing initiatives to increase customer activity and loyalty.

③ AI-powered churn prediction with SMS alerts

An end-to-end AI-powered retention system. The ensemble machine learning model, achieving a **69.70% accuracy** , identifies customers most likely to leave. An AI agent then automatically generates a personalized SMS for each high-risk customer — delivered at the right moment, before they decide to switch.

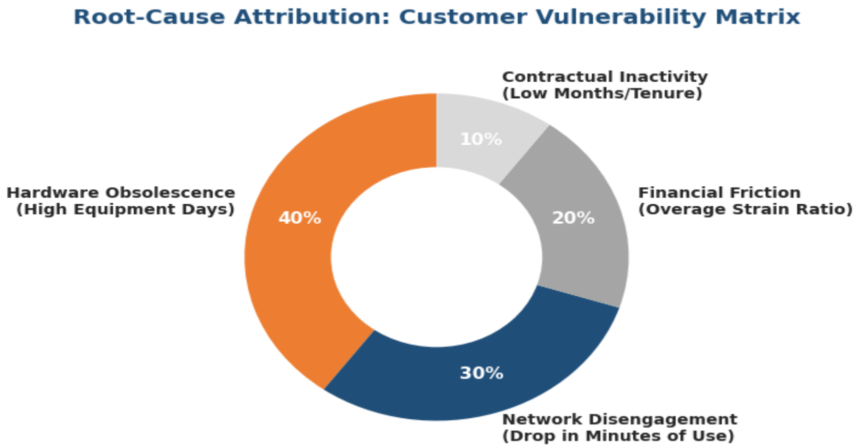
Business Proposal 1

Problems leading to customer churn :

Customers with older devices tend to incur higher overage charges and exceed their plan limits more frequently. This suggests that churn is driven not only by aging equipment but also by dissatisfaction arising from plan-device mismatches and increased billing costs

Root Cause Hypothesis
Customers using aging devices while frequently exceeding their service limits may perceive lower value from their current plan and become more susceptible to competitor offers

Customers with aging devices and high overage usage are significantly more likely to churn, indicating dissatisfaction with their current device-plan experience.

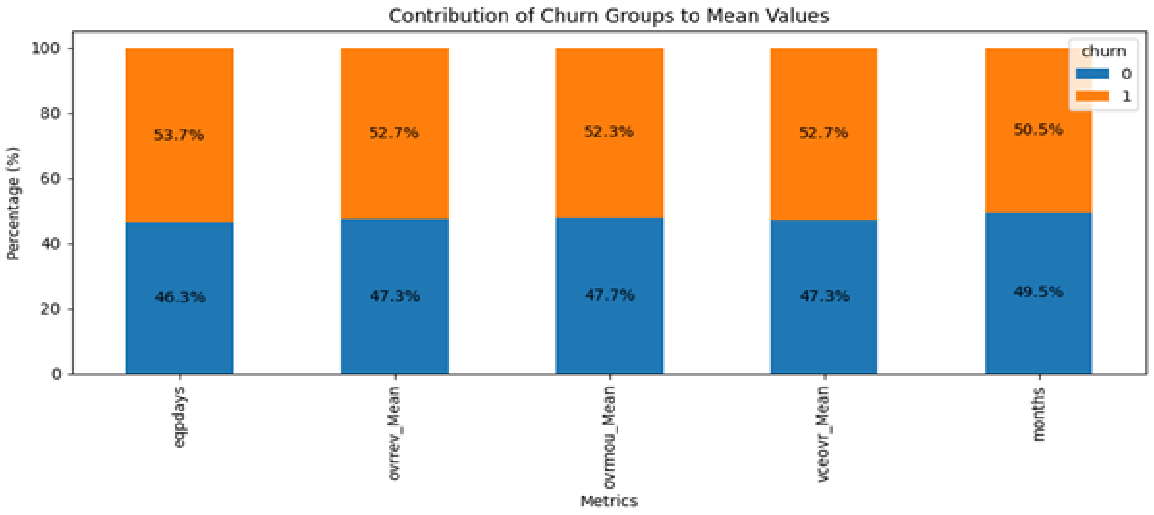


Company A's core revenue is threatened by high-value subscriber attrition driven by 'Hardware Obsolescence' and 'Bill Shock

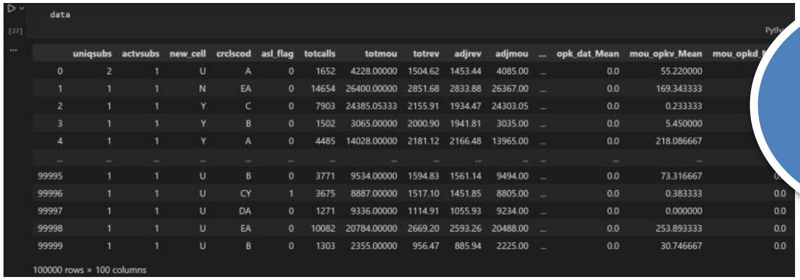
Proposed Solutions :

Launch a **Device Upgrade + Smart Plan Migration Program**:
For high-risk customers:
•Handset upgrade offers
•Trade-in incentives
•Personalized plan recommendations
•Bundled voice/data packages

Analysis shows churners not only have older devices but also consistently incur higher overage charges and minutes. Upgrading customers to more suitable device-plan combinations can improve perceived value and reduce the likelihood of switching providers.



Overview of the Machine Learning Model and the data insights



data

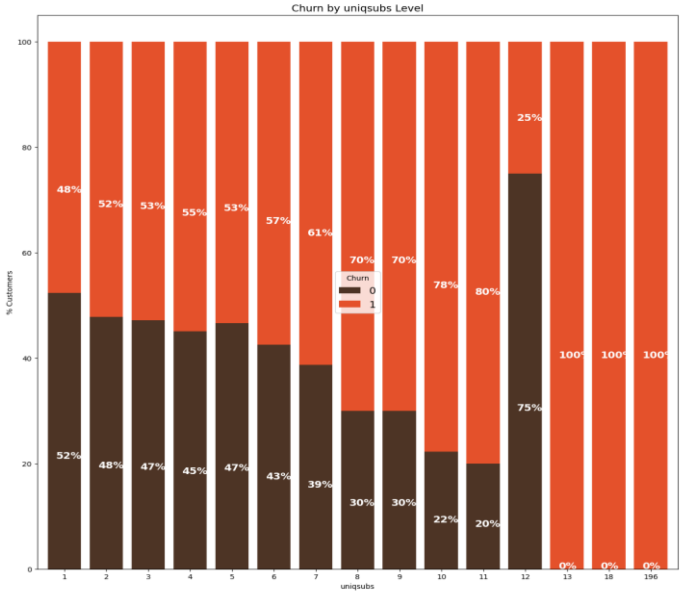
	uniqsubs	actvsubs	new_cell	crclscod	asl_flag	totcalls	totmou	totrev	adjrev	adjmou	...	opk_dat_Mean	mou_opkv_Mean	mou_opkd_
0	2	1	U	A	0	1652	4228.00000	1504.62	1453.44	4085.00	...	0.0	55.220000	
1	1	1	N	EA	0	14654	26400.00000	2851.68	2833.88	26367.00	...	0.0	169.343333	
2	1	1	Y	C	0	7903	24385.05333	2155.91	1934.47	24303.05	...	0.0	0.233333	
3	1	1	Y	B	0	1502	3065.00000	2000.90	1941.81	3035.00	...	0.0	5.450000	
4	1	1	Y	A	0	4485	14028.00000	2181.12	2166.48	13965.00	...	0.0	218.086667	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
99995	1	1	U	B	0	3771	9534.00000	1594.83	1561.14	9494.00	...	0.0	73.316667	
99996	1	1	U	CY	1	3675	8887.00000	1517.10	1451.85	8805.00	...	0.0	0.383333	0.0
99997	1	1	U	DA	0	1271	9336.00000	1114.91	1055.93	9234.00	...	0.0	0.000000	0.0
99998	1	1	U	EA	0	10082	20784.00000	2669.20	2593.26	20488.00	...	0.0	253.893333	0.0
99999	1	1	U	B	0	1303	2355.00000	956.47	885.94	2225.00	...	0.0	30.746667	0.0

100000 rows × 100 columns

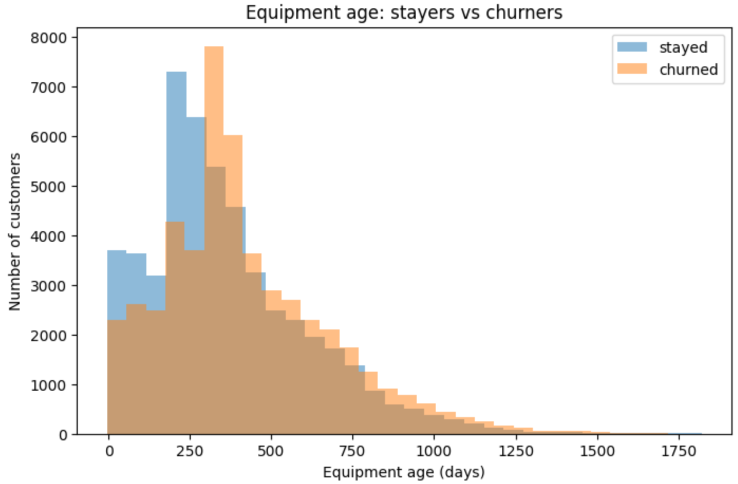
Approximately data of 100,000 customers was accumulated

Data contents
- Details of customers such as dwelling, family information,churn,etc
- Customers information such as credit score, calls dropped,etc
- Monthly usage scores, total revenue
- Unique subscribers in house

Data Visualization

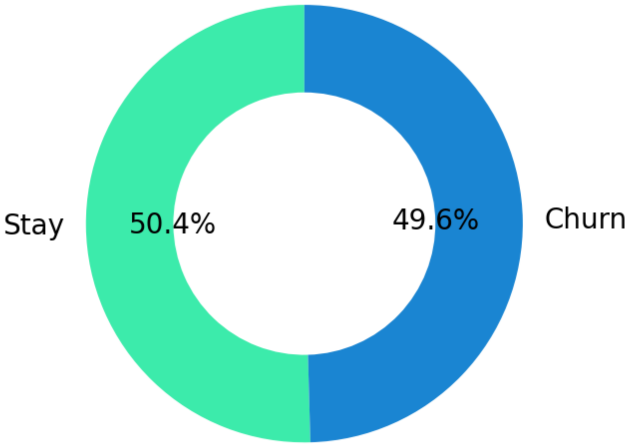


Customers with higher numbers of unique subscriptions (uniqsubs) exhibit significantly higher churn rates, making subscription complexity a strong indicator of customer attrition risk.



Customers with older equipment are more likely to churn, indicating that device age is a strong predictor of customer attrition.

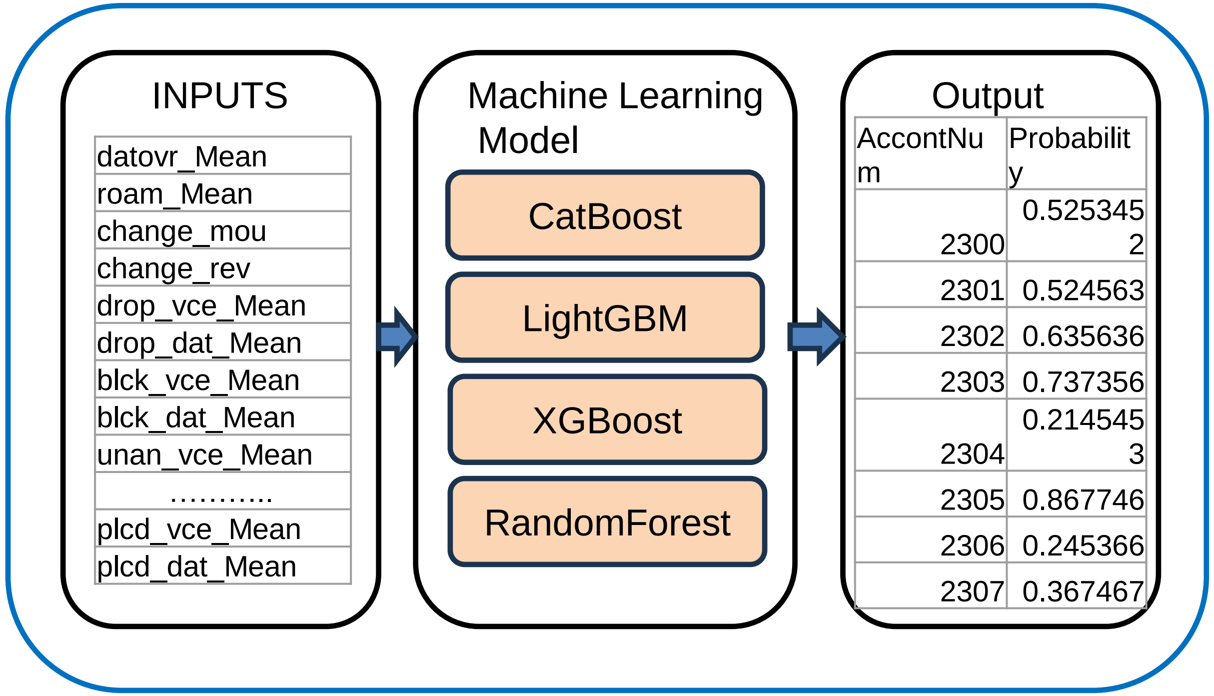
CUSTOMER CHURN DISTRIBUTION



IT IS VERY DIFFICULT TO ANALYZE THE CHURNERS WITH INDIVIDUAL FEATURES , LEADING TO OVER 100+ GRAPHS

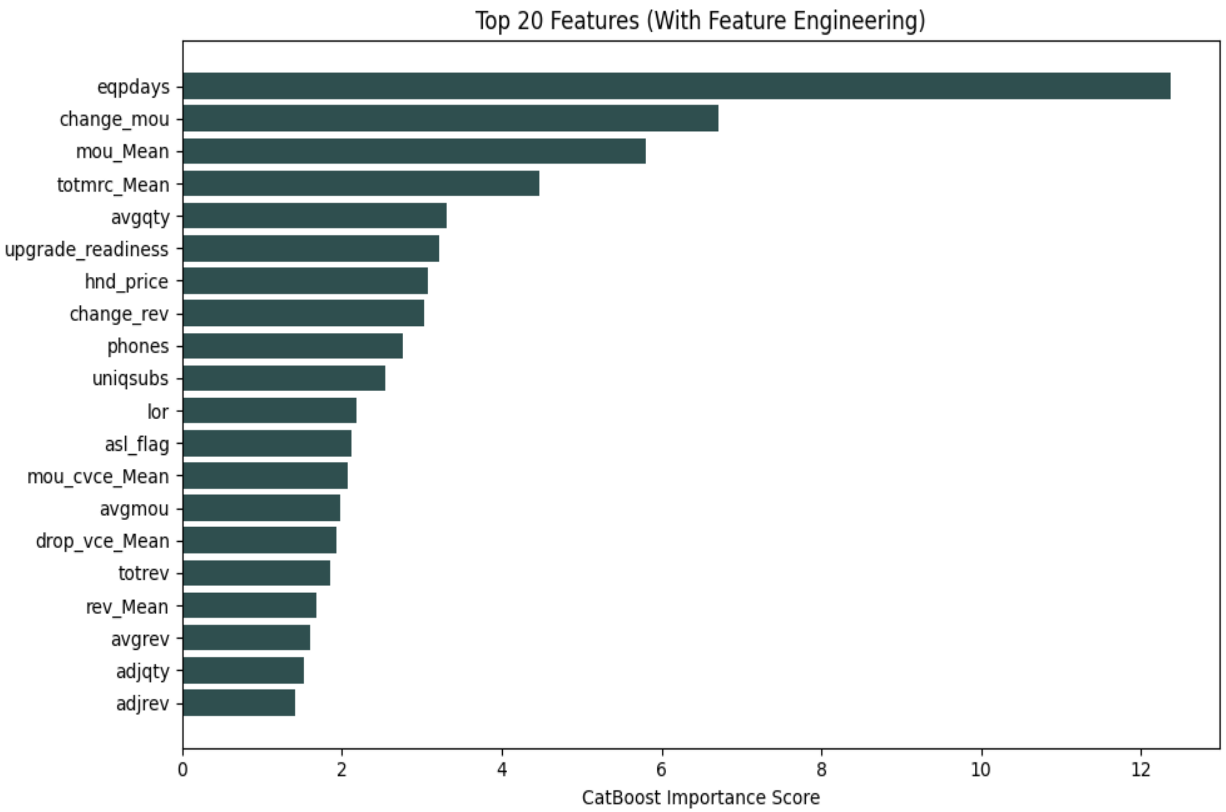
ADVANCE FEATURE ENGINEERING WITH MODEL SELECTING THE MOST PROMINET REASONS FOR CHURNING OF CUSTOMERS MAKING ANALYSIS EASIER AND QUICKER

WORKING OF THE MODEL



- **CatBoost (AUC = 0.6970):** Chosen for handling categorical telecom features smoothly, confirming plan–device mismatch drives churn.
- **LightGBM (AUC = 0.6949):** Selected for speed and efficiency on large datasets, revealed device age + overage revenue as top churn predictors.
- **XGBoost (AUC = 0.6948):** Used for robust tabular modeling, highlighted low usage and high billing shocks as churn signals.
- **Random Forest (AUC = 0.6587):** Baseline ensemble for comparison, showed weaker accuracy but useful for feature importance checks.
- **Blended Ensemble AUC: 0.6977**
- **Accuracy: 0.6416**

Important features for prediction



Features derived from processed data and statistical measures are important for AI predictions, indicating that the AI takes a different approach from human decision-making. Using both these features leads to a more accurate model and accurate predictions

Many of the features important for prediction are those related to values from external sources.

Revenue Maximization System through Churn Prediction

Problem

Customer churn directly impacts revenue and customer lifetime value.
Acquiring a new customer is significantly more expensive than retaining an existing one.
For a provider with 1 million customers at a $50 monthly Average Revenue Per User (ARPU), a 20% churn rate translates to $120 million in lost revenue each year.*
Using machine learning, the company can proactively identify customers likely to churn and intervene before they leave.

Customer Data
↓
ML Churn Prediction Model
(AUC = 0.69)
↓
High-Risk Customers Identified
↓
Retention Actions Triggered
↓
Reduced Churn
↓
Revenue Retained

Segment 1: Older Device Users

Evidence:

- High eqpdays
- High churn

Action:

- Device upgrade offers
- Trade-in discounts

Segment 2: High Overage Users

Evidence:

- High ovrrev_Mean
- High ovrmou_Mean

Action:

- Personalized plan migration
- Unlimited bundles

Segment 3: Low Usage Customers

Evidence:

- Lower usage linked with higher churn

Action:

- Engagement campaigns
- Loyalty rewards

Example of Application 1

Attribute	Value
Customer ID	C10245
Device Age (eqpdays)	420 days
Handset Price	$53
Overage Minutes	High
Overage Revenue	High
Monthly Usage	Low
Predicted Churn Probability	87%

```
Sample:
       Customer_ID  churn  churn_prob
13710      1013711      1    0.550458
46836      1046837      1    0.659465
14490      1014491      1    0.655947
97963      1097964      1    0.731558
53674      1053675      1    0.617459
```

Model Prediction
•Predicted Outcome = Churn
•Probability = 87%
•Risk Level = High

Model Prediction

- Predicted Outcome = **Churn**
- Probability = **87%**
- Risk Level = **High**

Expected Outcome

Without intervention:

- Customer likely leaves competitor.

With intervention:

- Customer receives better device + suitable plan.

Result:

- Retention probability increases.
- Revenue is preserved

Automated Action
Since churn probability > 70%:
System automatically recommends
✓ Device Upgrade Offer
✓ Smart Plan Migration
✓ Loyalty Discount
✓ Personalized Retention Call

Revenue Prediction by PCME

Assumptions

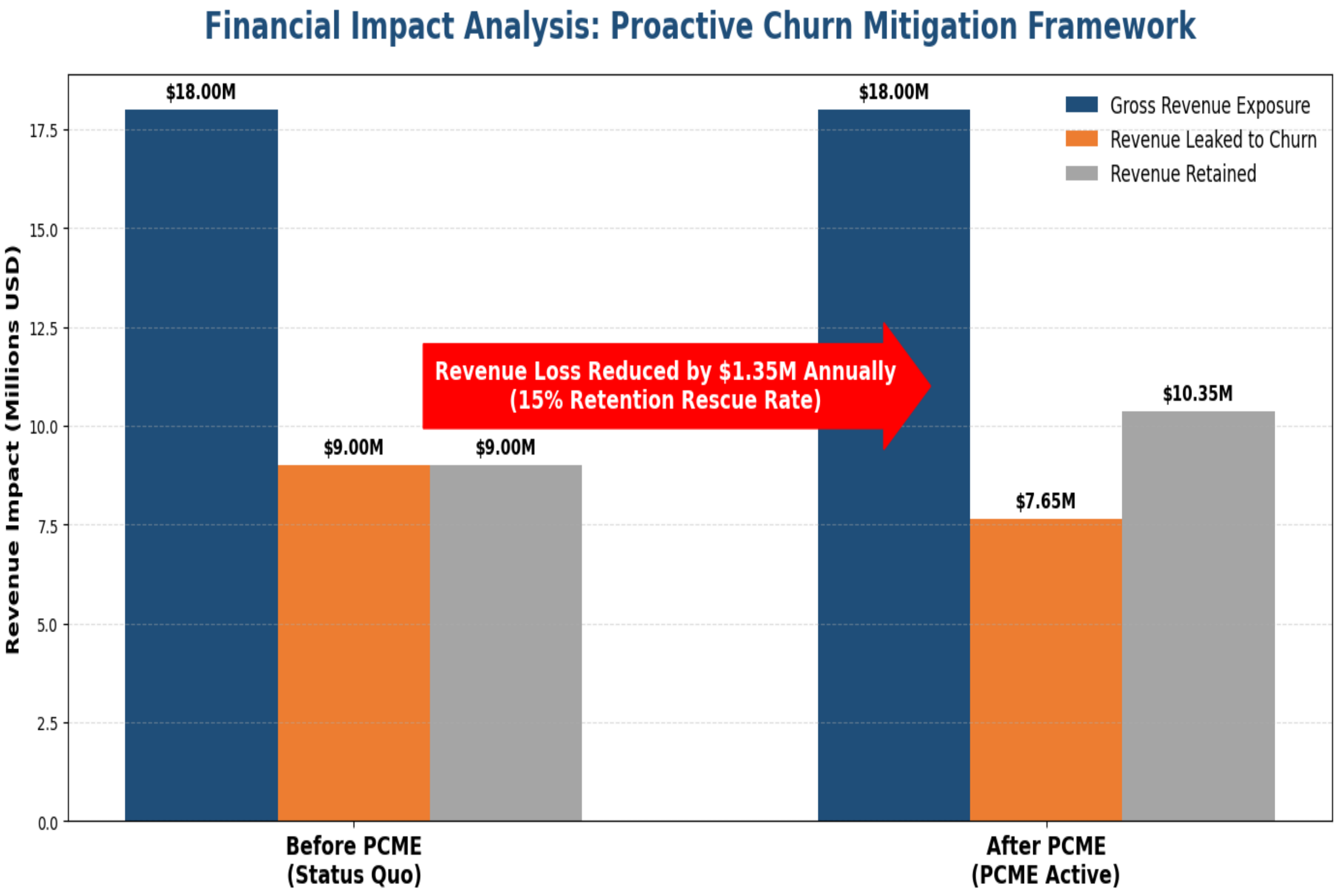
- Total Customers = 100,000
- Churn Rate = 50%
- ARPU = $15/month**
- Retention Rescue Rate = 15%
- Simulation Period = 12 months

Revenue Base = 100k × $15 × 12 = $18.0M
Revenue Saved = $9.0M × 15% = $1.35M

↓ $1.35M Saved

Key Insight:
Annual revenue leakage reduced by approximately $1.35M through proactive retention.

Key Insight
By identifying high-risk customers before they leave and triggering targeted retention actions, the Proactive Churn Mitigation Engine (PCME) can potentially reduce annual revenue leakage by **$1.35 million**, assuming a 15% retention rescue rate.



Business Proposal (2) Churn Customer Trend Analysis System current problems and solutions

Business Problem
Telecom companies often react to customer churn after it occurs, resulting in lost revenue and missed retention opportunities.
With thousands of customers and numerous behavioral attributes, identifying emerging churn trends manually is difficult

Key Findings from Analysis (Problems)

Device-Related Trends
- **Older devices (eqpdays) show higher churn.**
- **Lower handset prices (hnd_price) are associated with higher churn.**

Usage Trends
- **Lower Minutes of Use (mou_Mean) correspond to higher churn.**
- **Lower Voice Usage (mou_cvce_Mean) corresponds to higher churn.**
- **Lower Completed Calls (complete_Mean) correspond to higher churn.**

Billing Trends
- **Higher overage revenue (ovrrev_Mean) corresponds to higher churn.**
- **Higher overage minutes (ovrmou_Mean) correspond to higher churn.**

Existing Data
+
Foreign Exchange Data

Input

Machine Learning

Lending adapted to financial markets and economic trends becomes possible.

In addition, model updates can be implemented to improve the accuracy of lending decisions.

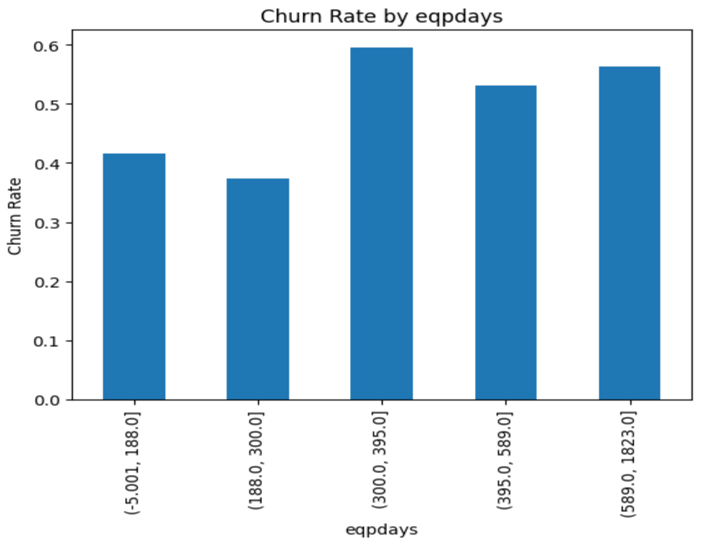
Proposed solutions

✅ **By detecting changes in churn trends with high sensitivity, the company can respond to customer needs more quickly than competitors, reducing customer attrition and improving retention rates.**
✅ **Recommends personalized retention strategies for each customer based on historical behavior, usage patterns, device characteristics, and churn risk predictions generated by the model.**
✅ **Enables marketing and customer success teams to focus retention campaigns on high-risk customer segments, optimizing resource allocation and improving campaign effectiveness.**
✅ **Provides continuous monitoring of customer behavior trends, allowing management to identify emerging churn patterns before they become significant revenue risks.**

Use Case 2: Issues Visualized through Default Trend Analysis

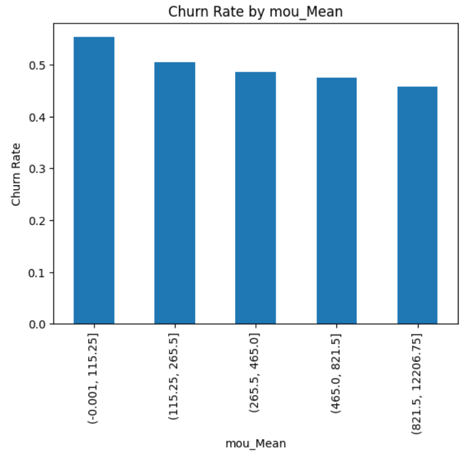
1. Days Since Equipment/Phone Activation [EPQDAYS]
from the graph ,Customers with older devices tend to exhibit higher churn rates. Churn increases significantly after approximately 300 days, suggesting that customers nearing the end of their device lifecycle are more likely to switch providers.

Business Action:
Target customers with older devices using upgrade offers and retention campaigns.



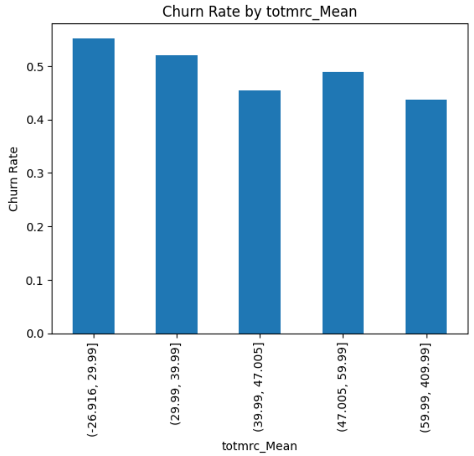
2. Average Minutes of Use [MOU_MEAN]
Churn steadily decreases as customer usage increases. Low-usage customers are substantially more likely to leave than highly engaged customers.

Business Action:
Increase engagement through bundled services, promotional usage packs, and loyalty incentives.



3. Average Monthly Revenue Charge [TOTMRC_MEAN]
Customers with lower monthly revenue charges show higher churn rates, while customers contributing higher monthly revenue tend to be more loyal

Business Action:
Introduce value-added plans and personalized offers for low-revenue customers to improve retention.



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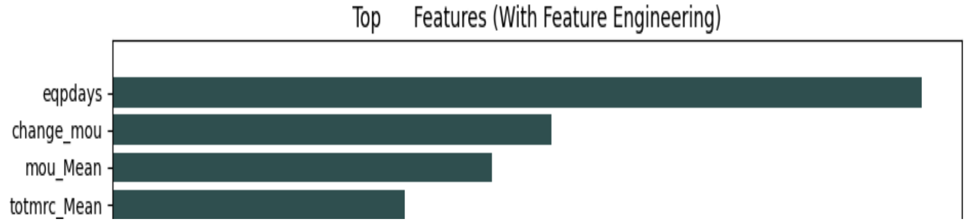

3)AI-Powered Churn Retention System

"This proposal focuses on one goal: delivering a solution that works in the real world."

49.6% churn rate
49,562 at-risk customers

Dropped Calls | **Bill Overcharges** | **Usage Dropping**

Problem

Telecom companies lose thousands of customers every month to churn. The real cost isn't just lost revenue — it's the inability to act before a customer decides to leave. Traditional systems only detect churn after it happens, when it's already too late.

Solution

An end-to-end AI pipeline that identifies high-risk customers using an ensemble ML model, then automatically triggers a personalized SMS — crafted by an AI agent — at exactly the right moment to retain them.

The Process

1. Customer behavior data is recorded in the database
2. Ensemble model predicts churn probability for each customer
3. High-risk customers are filtered (top decile)
4. AI agent generates a personalized SMS per customer
5. Message is delivered to the customer's phone instantly

The Pipeline

Database → **ML Model** → **AI Agent** → **Customer Phone**

- **Database**: Customer data stored & updated
- **ML Model**: Our ensembled model predicts churn probability
- **AI Agent**: Groq receives the problems and generates personalized SMS
- **Customer Phone**: SMS delivered at right moment

Sample SMS: *"Dear Customer, you may be eligible for our Gold Plan which caps voice overcharges. Reply YES to switch & save. T&C apply."*

Results, Revenue Impact & Roadmap

High-risk customers identified	**49,562**
SMS retention rate (industry avg)	**10%**
Customers retained	**4,956**
Avg revenue per user (ARPU)	**$57.91**
Revenue saved per month	**$287,002**
Revenue saved per year	**$3,444,024**

✅ Phase 1 — Complete

- Predict IF a customer will churn
- Ensemble model (CatBoost + XGBoost + LightGBM + Random forest)
- This predicted Values are then sent to Groq to analyze.
- AI agent auto-generates personalized SMS via Groq using the predicted value
- Predict WHEN they will churn (7 / 30 / 90 days)
- Survival Analysis: Cox Proportional Hazard Model
- Send offer at exactly the right moment — before they decide

🔮 Phase 2 — PDCA CYCLE

•**Plan:** Frame the churn hypothesis.
•**Do:** Run the AI-Powered SMS Retention Agent.
•**Check:** Use the testing and **Revenue Impact Analysis**.
•**Act:** Automate the process through proposed API integration

References

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Quotation

Sale of AI Churn Prediction & SMS Retention System

Content:

•Automatically identifies customers with high likelihood of churn based on ensemble ML model (CatBoost + LightGBM + XGBoost)
•System automatically generates personalized AI-driven SMS offers for each high-risk customer at the optimal moment
•Analyzes behavioral trends (usage drop, overcharges, roaming) to enable early detection and proactive retention

Price:

We are offering the complete AI-based churn retention system for a total price of
$688,833/year
For a telecom operator of 100,000 customers, an annual revenue recovery of approximately $1.35M to $3.4M is expected. Taking into account its strong **be provided free of charge.** ROI of 400% and low operational cost of just $377/campaign, we have set the price at this level. Also, **for the first year after signing, consulting services will**

Consulting Contract

Content:

•Training and onboarding for in-house data and marketing teams
•Retraining of prediction models on updated customer data
•Development of Phase 2 survival analysis (predicting WHEN customers churn)
•Proposals for new AI-driven business retention strategies

Price:

The monthly contract is set at **$7,500/month**, and the annual contract at **$82,000/year.**
Regarding new business proposals introduced above, all consulting engagements can be undertaken based on the pricing indicated here as the standard fee. (Additional performance-based compensation may be applied depending on outcomes.)